

3.6 Partial correlation

Two variables can be strongly correlated only because both depend on a third. Example (Holding a third variable fixed) showed a positive association between Y_1 and Y_2 turning negative once Y_3 is held fixed. The conditional covariance matrix $\Sigma_{11 \cdot 2}$ measures association that remains after conditioning, and its correlation form is the partial correlation.

3.6.1 Definition and interpretations

Definition 3.6.1 — Partial correlation

Let $\mathbf{Y} = (\mathbf{Y}_1^\top, \mathbf{Y}_2^\top)^\top$ be a random vector with covariance matrix Σ , partitioned as in Section 3.5, and write $\Sigma_{11 \cdot 2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} = (\sigma_{ij \cdot 2})$. The **partial covariance** of Y_i and Y_j (two entries of $\mathbf{Y}_1$) given $\mathbf{Y}_2$ is $\sigma_{ij \cdot 2}$, and their **partial correlation** given $\mathbf{Y}_2$ is

$$\rho_{ij \cdot 2} = \frac{\sigma_{ij \cdot 2}}{\sqrt{\sigma_{ii \cdot 2} \sigma_{jj \cdot 2}}},$$

provided $\sigma_{ii \cdot 2}$ and $\sigma_{jj \cdot 2}$ are positive. When the conditioning variables are $Y_k, Y_l, \dots$ we also write $\rho_{ij \cdot kl \dots}$. The matrix of partial correlations is $\mathbf{D}^{-1/2} \Sigma_{11 \cdot 2} \mathbf{D}^{-1/2}$ with $\mathbf{D} = \text{diag}(\Sigma_{11 \cdot 2})$, the correlation matrix (Definition 2.4.1) built from $\Sigma_{11 \cdot 2}$.

The definition uses only Σ , so partial correlations exist for any random vector with finite second moments. The next result gives them two meanings, one that always applies and one that needs normality.

Proposition 3.6.1

In the setting of Definition 3.6.1:

- (a) $\Sigma_{11 \cdot 2}$ is the covariance matrix of the prediction errors $\mathbf{W} = \mathbf{Y}_1 - \boldsymbol{\mu}_1 - \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{Y}_2 - \boldsymbol{\mu}_2)$, and $\mathbf{W}$ is uncorrelated with $\mathbf{Y}_2$. If Σ_{22} is positive definite, the entries of $\mathbf{W}$ are the errors of the best linear predictors $L(Y_i | \mathbf{Y}_2)$, so $\rho_{ij \cdot 2}$ is the correlation between $Y_i - L(Y_i | \mathbf{Y}_2)$ and $Y_j - L(Y_j | \mathbf{Y}_2)$.
- (b) If $\mathbf{Y}$ is jointly normal, $\rho_{ij \cdot 2}$ is the correlation of Y_i and Y_j in their conditional distribution given $\mathbf{Y}_2 = \mathbf{y}_2$, the same for every $\mathbf{y}_2$. Moreover $\rho_{ij \cdot 2} = 0$ iff Y_i and Y_j are conditionally independent given $\mathbf{Y}_2$.

Proof.

- (a) This is Theorem 2.5.2, restated: $\text{Cov}(\mathbf{W}, \mathbf{Y}_2) = \mathbf{0}$ and $\text{Cov}(\mathbf{W}) = \Sigma_{11 \cdot 2}$ are the moment calculations in the proof of Theorem 3.5.2(a), which use no normality. When Σ_{22} is positive definite, the i th entry of $\boldsymbol{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{Y}_2 - \boldsymbol{\mu}_2)$ is $L(Y_i | \mathbf{Y}_2)$ by Theorem 6.11.1.
- (b) By Theorem 3.5.2, the conditional distribution of (Y_i, Y_j) given $\mathbf{Y}_2 = \mathbf{y}_2$ is bivariate normal with covariance matrix the corresponding 2×2 block of $\Sigma_{11 \cdot 2}$. Its correlation is $\rho_{ij \cdot 2}$, and by Theorem 3.4.1 applied to this conditional normal distribution, the two are independent iff the off-diagonal entry is zero.

Part (a) is the definition to remember: *a partial correlation is the correlation of two residuals*. Remove from each variable the part that is linearly predictable from the conditioning variables, and correlate what is left. This is the population version of the Frisch–Waugh–Lovell construction (Theorem 6.6.2).

3.6.2 Computing partial correlations

Two routes avoid computing $\Sigma_{11 \cdot 2}$ in full.

With a single conditioning variable, $\Sigma_{11 \cdot 2}$ is a 2×2 matrix with entries $\sigma_{ij} - \sigma_{i3}\sigma_{j3}/\sigma_{33}$, and in correlation units this is the familiar three-variable formula of Exercise (B1),

$$\rho_{12 \cdot 3} = \frac{\rho_{12} - \rho_{13}\rho_{23}}{\sqrt{(1 - \rho_{13}^2)(1 - \rho_{23}^2)}}, \quad |\rho_{13}| < 1, \quad |\rho_{23}| < 1. \quad (3.6.1)$$

The residual interpretation shows that the same formula can be applied one conditioning variable at a time.

Proposition 3.6.2 — Recursion

Let S be a set of indices and $k \notin S \cup \{1, 2\}$ one more. Suppose $\sigma_{11 \cdot S}$, $\sigma_{22 \cdot S}$ and $\sigma_{kk \cdot S}$ are positive and $|\rho_{1k \cdot S}| < 1$, $|\rho_{2k \cdot S}| < 1$. Then

$$\sigma_{12 \cdot Sk} = \sigma_{12 \cdot S} - \frac{\sigma_{1k \cdot S} \sigma_{2k \cdot S}}{\sigma_{kk \cdot S}}, \quad \rho_{12 \cdot Sk} = \frac{\rho_{12 \cdot S} - \rho_{1k \cdot S} \rho_{2k \cdot S}}{\sqrt{(1 - \rho_{1k \cdot S}^2)(1 - \rho_{2k \cdot S}^2)}}.$$

With S empty this is [Eq. 3.6.1](#).

Proof. No normality is needed. For $i \in \{1, 2, k\}$ let e_i be the error of the best linear predictor of Y_i from $\mathbf{Y}_S$, as in [Proposition 3.6.1\(a\)](#). Then each e_i has mean zero, is uncorrelated with $\mathbf{Y}_S$, and $\text{Cov}(e_i, e_j) = \sigma_{ij \cdot S}$. Put $r_i = e_i - c_i e_k$ with $c_i = \sigma_{ik \cdot S}/\sigma_{kk \cdot S}$, for $i = 1, 2$.

We claim that r_i is the error of the best linear predictor of Y_i from $(\mathbf{Y}_S, Y_k)$. First, $Y_i - r_i$ is an affine function of $(\mathbf{Y}_S, Y_k)$, because e_i and e_k differ from Y_i and Y_k by affine functions of $\mathbf{Y}_S$. Second, r_i has mean zero and is uncorrelated with $\mathbf{Y}_S$ and with e_k , by the choice of c_i , hence with $Y_k = e_k + (\text{affine in } \mathbf{Y}_S)$. Now let r'_i be the error from [Proposition 3.6.1\(a\)](#) for predicting Y_i from $(\mathbf{Y}_S, Y_k)$, which has the same two properties. The difference $d = r_i - r'_i$ is an affine function of $(\mathbf{Y}_S, Y_k)$ with mean zero and is uncorrelated with $(\mathbf{Y}_S, Y_k)$, so $\text{Var}(d) = \text{Cov}(d, d) = 0$. Thus $r_i = r'_i$ with probability one.

Therefore $\sigma_{12 \cdot Sk} = \text{Cov}(r_1, r_2) = \sigma_{12 \cdot S} - c_2 \sigma_{1k \cdot S} - c_1 \sigma_{2k \cdot S} + c_1 c_2 \sigma_{kk \cdot S}$, which simplifies to the first formula, and similarly $\sigma_{ii \cdot Sk} = \sigma_{ii \cdot S}(1 - \rho_{ik \cdot S}^2)$, which is positive by hypothesis. Dividing $\sigma_{12 \cdot Sk}$ by $\sqrt{\sigma_{11 \cdot Sk} \sigma_{22 \cdot Sk}}$ and then numerator and denominator by $\sqrt{\sigma_{11 \cdot S} \sigma_{22 \cdot S}}$ gives the second.

Proposition 3.6.3 — Partial correlations from the precision matrix

Let Σ be positive definite and $\Omega = \Sigma^{-1} = (\omega_{ij})$. The partial correlation of Y_i and Y_j given all the other entries of $\mathbf{Y}$ is

$$\rho_{ij \cdot \text{rest}} = -\frac{\omega_{ij}}{\sqrt{\omega_{ii}\omega_{jj}}},$$

and the conditional variance of Y_i given all the others is $1/\omega_{ii}$.

Proof. Permute $\mathbf{Y}$ so that (Y_i, Y_j) comes first. By Theorem 3.5.2(c) (a statement about matrices, valid whether or not $\mathbf{Y}$ is normal), the leading 2×2 block of Ω is $\Sigma_{11 \cdot 2}^{-1}$. Inverting,

$$\Sigma_{11 \cdot 2} = \begin{pmatrix} \omega_{ii} & \omega_{ij} \\ \omega_{ij} & \omega_{jj} \end{pmatrix}^{-1} = \frac{1}{\omega_{ii}\omega_{jj} - \omega_{ij}^2} \begin{pmatrix} \omega_{jj} & -\omega_{ij} \\ -\omega_{ij} & \omega_{ii} \end{pmatrix},$$

whose correlation is $-\omega_{ij}/\sqrt{\omega_{ii}\omega_{jj}}$. The same argument with a 1×1 leading block gives the conditional variance.

Proposition 3.6.3 has a striking consequence for normal vectors: *a zero entry ω_{ij} of the inverse covariance matrix means that Y_i and Y_j are conditionally independent given all the other variables.* Zeros in Σ describe marginal independence, zeros in Σ^{-1} describe conditional independence. The pattern of zeros in Σ^{-1} defines a graph on the variables, which is the starting point of Gaussian graphical models (Dempster 1972; Lauritzen 1996).

Example. (The trivariate example, continued)

For Σ of Example (Holding a third variable fixed), the listing gives the covariance of (Y_1, Y_2) given Y_3 as $\begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$. So $\rho_{12} = 0.289$ while $\rho_{12 \cdot 3} = -0.707$. The script confirms that Eq. 3.6.1, Proposition 3.6.3, and the correlation of simulated residuals all give this value.

3.6.3 Sample partial correlations

For data, replace Σ by the sample covariance matrix. Because the empirical distribution of the rows is a probability distribution in its own right (Section 6.11), all the identities above hold exactly for the sample versions. In particular, for two variables $\mathbf{y}, \mathbf{x}$ and conditioning variables collected with an intercept in $\mathbf{Z}$, the following numbers are *identical*:

- (1) the correlation of the residuals of $\mathbf{y}$ and of $\mathbf{x}$ after least squares regression on $\mathbf{Z}$;
- (2) the recursion Eq. 3.6.1 applied to sample correlations (one conditioning variable) or repeatedly (several);
- (3) $-w_{12}/\sqrt{w_{11}w_{22}}$ computed from the inverse of the sample covariance or correlation matrix

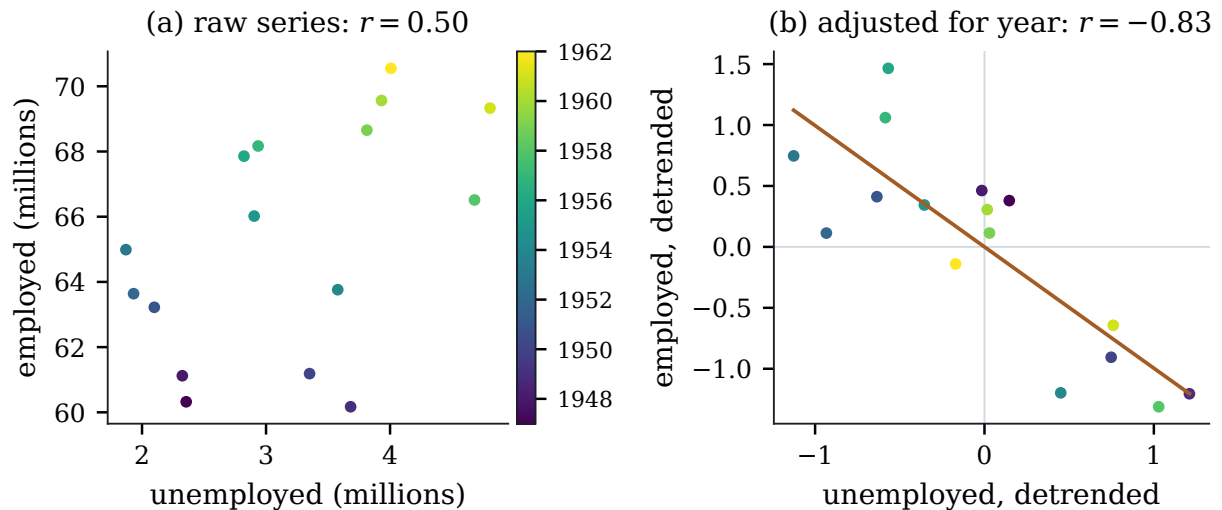


Figure 3.6.1: Employment and unemployment in the United States, 1947–1962, coloured by year. (a) The raw series are positively correlated because both trend upwards. (b) After removing the linear trend in year from each, the residuals are strongly negatively correlated. Their correlation is the partial correlation given year, and the fitted slope is the multiple-regression coefficient.

of $(\mathbf{y}, \mathbf{x}, \text{conditioning variables})$;

- (4) $\text{sign}(t)\sqrt{t^2/(t^2 + n - p)}$, where t is the t statistic of $\mathbf{x}$ in the regression of $\mathbf{y}$ on $[\mathbf{Z}, \mathbf{x}]$ with p columns (Exercise (B3)).

Example. (Employment and unemployment)

Longley’s macroeconomic series record, among other things, total employment and the number unemployed in the United States for the 16 years 1947–1962. Their correlation is 0.502: years with more people employed had more people unemployed. Both series grew with the population. Their correlations with the calendar year are 0.971 and 0.668. Adjusting for year, the partial correlation is -0.829 . Once the common trend is removed, a year with unemployment above its trend is a year with employment below its trend, as one would expect. Figure 3.6.1 shows both scatterplots. The listing computes the partial correlation in all four ways above. They agree to ten decimal places. The slope in panel (b), -0.996 , is the coefficient of unemployment in the regression of employment on unemployment and year, as Theorem 6.6.2 requires. Its t statistic is -5.34 .

Removing a linear time trend before correlating two series is the problem that Frisch and Waugh (1933) solved, and it is a special case of partial correlation.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.longley.load_pandas().data

emp = data["TOTEMP"].to_numpy() / 1000      # thousands -> millions of persons
unemp = data["UNEMP"].to_numpy() / 1000
year = data["YEAR"].to_numpy()
n = len(emp)
Z = np.column_stack([np.ones(n), year])      # the variables we adjust for

def resid(v, Z):
    coef, *_ = np.linalg.lstsq(Z, v, rcond=None)
```

```

    return v - Z @ coef

# 1. correlation of residuals (detrended series)
r_resid = np.corrcoef(resid(emp, Z), resid(unemp, Z))[0, 1]

# 2. the one-variable recursion from ordinary correlations
R = np.corrcoef(np.vstack([emp, unemp, year]))
r12, r13, r23 = R[0, 1], R[0, 2], R[1, 2]
r_formula = (r12 - r13 * r23) / np.sqrt((1 - r13**2) * (1 - r23**2))

# 3. from the inverse of the correlation (or covariance) matrix
W = np.linalg.inv(R)
r_precision = -W[0, 1] / np.sqrt(W[0, 0] * W[1, 1])

# 4. from the t statistic of unemployment in the regression of emp on (1, unemp,
    year)
fit = sm.OLS(emp, np.column_stack([Z, unemp])).fit()
t = fit.tvalues[2]
r_t = np.sign(t) * np.sqrt(t**2 / (t**2 + n - 3))

print(f"marginal r = {r12:.4f}")
print(f"partial r:  {r_resid:.10f} {r_formula:.10f} {r_precision:.10f} {r_t:.10f}")

```

[Run this code online](#)

Remark (Adjusting is not always right). A partial correlation answers a precise question: how are the parts of two variables not linearly explained by $\mathbf{Y}_2$ related? Whether that is the *right* question depends on what $\mathbf{Y}_2$ is. Adjusting for a common cause, as the year stands in for population growth here, removes a spurious association. Adjusting for a common effect can *create* an association between variables that are independent. [Chapter 25](#) treats the difference (Theorem 25.5.1). Under normality, the sampling distribution of a sample partial correlation given q variables is that of an ordinary sample correlation from $n - q$ observations, a result used for tests in [Chapter 14](#).

3.6.4 Exercises

3.6.4.1 A. Check your understanding

Exercise A1.

Show that $\rho_{12.3} = 0$ iff $\rho_{12} = \rho_{13}\rho_{23}$. Find $\rho_{12.3}$ when $\rho_{12} = 0$ and $\rho_{13} = \rho_{23} = 0.6$, and interpret.

3.6.4.2 B. Practice

Exercise B1.

Let X , Z and e be independent, with $X, Z \sim N(0, 1)$ and $e \sim N(0, \tau^2)$, and let $C = X + Z + e$. Show that $\rho_{XZ} = 0$ but $\rho_{XZ.C} = -1/(1 + \tau^2)$. What happens as $\tau \rightarrow 0$ and as $\tau \rightarrow \infty$?

Solution. The covariance matrix of (X, Z, C) is $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2+\tau^2 \end{pmatrix}$. Conditioning on C ,

$$\sigma_{XZ \cdot C} = 0 - \frac{1}{2 + \tau^2}, \quad \sigma_{XX \cdot C} = \sigma_{ZZ \cdot C} = 1 - \frac{1}{2 + \tau^2} = \frac{1 + \tau^2}{2 + \tau^2},$$

so $\rho_{XZ \cdot C} = -1/(1 + \tau^2)$. As $\tau \rightarrow 0$ it tends to -1 : if $X + Z$ is known exactly, a larger X forces a smaller Z . As $\tau \rightarrow \infty$ it tends to 0, since C carries little information about $X + Z$. Two independent causes become dependent once their common effect is held fixed.

Exercise B2.

Let Σ be positive definite. Show that the coefficient of Y_j in the best linear predictor of Y_1 from all the other entries is $-\omega_{1j}/\omega_{11}$. Deduce that $\rho_{1j \cdot \text{rest}}^2$ equals the product of the coefficient of Y_j for predicting Y_1 and the coefficient of Y_1 for predicting Y_j , each from all the other variables.

Solution. Put Y_1 first. By Exercise (B5) with a scalar first block, the coefficient vector of the best linear predictor of Y_1 from the rest is $-\Omega_{12}/\omega_{11}$, whose j th entry is $-\omega_{1j}/\omega_{11}$. Likewise the coefficient of Y_1 in the predictor of Y_j is $-\omega_{j1}/\omega_{jj}$. Their product is $\omega_{1j}^2/(\omega_{11}\omega_{jj})$, which is $\rho_{1j \cdot \text{rest}}^2$ by [Proposition 3.6.3](#).

Exercise B3.

Let $\mathbf{D}$ be an $n \times k$ data matrix whose columns are centred, and let $\mathbf{S} = \mathbf{D}^\top \mathbf{D}$. Show, using Theorem 6.6.2 and the partitioned inverse, that $-w_{12}/\sqrt{w_{11}w_{22}}$, computed from $\mathbf{S}^{-1} = (w_{ij})$, equals the correlation of the residuals of columns 1 and 2 after regression on the remaining columns.

Exercise B4.

Using the Longley data of [Example \(Employment and unemployment\)](#), compute the partial correlation of employment and unemployment given both year and GNP, in at least two of the four ways. Draw the corresponding residual scatterplot. Does adjusting for GNP as well as year change the conclusion?